Moving Tech AI Website

Deployment and Management Guide

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1. Overview

The purpose of this document is to help explain how to update the MovingTechAI codebase via Github as well as the redeployment of all four services on Render.

Service	Type	Technology	Folder
Public Website Frontend	Static Site	React (Create React App)	movingTechAIFrontend
Public Website Backend	Web Service	Python (FastAPI)	movingTechAIBackend
Admin Portal Frontend	Static Site	React (Vite)	movingTechAIAdminPortalFrontend
Admin Portal Backend	Web Service	Node.js (Express)	movingTechAIAdminPortalBackend

2. Prerequisites & Access

Ensure you have the following setup on your computer before accessing and updating the codebase:

Tool / Service	Purpose	How to get access
Github Account	Push code changes to the repository	Ask to be added as a collaborator to the Github repository
Render Account	Monitor deployments, manage env vars, restart services	On the basic free plan, only the admin account is allowed access to the environment. To allow other contributors you must upgrade the plan
Firebase console	View and edit the database information directly	Ask the lead developer to have your gmail added to the access list
Git	Clone the github repository and push changes locally	Install from git-scm.com
Node.js 18+/NPM	Run and test the code locally	Install from nodejs.org
Python 3.10+/PIP	Run and test the code locally	Install from python.org

3. Making Changes on Github

Important: Never commit the serviceAccountKey.json or any .env file to the Github. These files should be listed in the .gitignore, however, always confirm your commits before pushing. If you accidentally commit a secret, get a new key immediately and update the env variables in Render.

3.1 Option A, Edit Directly on Github (Small edits)

This method requires no software installation and will work to add comments or small tweaks to the code.

1. Open the Github repo in your browser.
2. Navigate to the file you want to update.
3. Click on the "Edit" pencil icon in the top right of the file viewer.
4. Make your changes.
5. Click "Commit Changes".
6. Add a commit message and click "Commit Changes"

After updating the code ensure you check the Render environment to redeploy the code (See section 4.1)

3.2 Option B, Clone and Edit Locally (Larger changes)

This method gives full control over the coding IDE and allows for testing the code locally before publishing changes. This is the recommended approach for large scale changes to the codebase

1. Clone the repository to a folder.
2. Create a branch for your work.
3. Make your changes in the preferred IDE.
4. Test your changes locally.
5. Stage the commit and push to Github.
6. Create a merge request into the main branch after reviewing the code.

After updating the code ensure you check the Render environment to redeploy the code (See section 4.1)

4. Starting & Monitoring Deployments on Render

After pushing your code to Github, Render should automatically rebuild and deploy the new code changes, however, if this doesn't occur you can follow these steps:

4.1 Manual Redeployment on Render

1. Go to the Render Dashboard.
2. Select the "MovingTechAI Website Redesign" project.
3. Click on the service you need to update.
4. In the "Events" tab, check the most recent "Deploy live".
    a. If the deploy shows your most recent commit message, then the system automatically rebuilt and deployed the code.
    b. If the deploy shows as a previous commit message proceed to step 5.
5. Click "Manual Deploy" and "Deploy latest commit".

    a. You will see a full deploy log as it begins the rebuild and deployment. Once it's finished the status will show up top as either green for live or red for failure.
    b. If the deploy fails, read the log for the error messages and proceed accordingly.

4.2 Viewing Deployment Status on Render

1. Go to the Render Dashboard.
2. Select the "MovingTechAI Website Redesign" project.
3. The "Deployment Status" will show right next to the service.
4. Click on the service and go to the "Events" tab for a detailed breakdown of the current deployment.
    a. Click on the word "Deploy" for the given deployment to see the full deployment log.

5. Managing Environment Variables on Render

Environment Variables hold sensitive config data like API keys, passwords, credentials, etc. These should never be hard coded into any file, therefore, in each Render service there is environment variables which can be dynamically updated in the dashboard and pulled into the code.

5.1 How to View or Edit Environment Variables

1. Go to the Render Dashboard.
2. Select the "MovingTechAI Website Redesign" project.
3. Select the service which holds your environment variable(s).
4. Click on the "Environment" tab on the left hand navbar.
    a. Click on the "Edit" button to edit the variables.
    b. Click on the eye icon next to the variable to see the raw value.

5.2 Required Environment Variables by Service

Public Website Frontend:

Variable Name	What it contains	Required?
REACT_APP_API_URL	The full HTTPS URL for the backend WebService for the public facing website.	Yes

Public Website Backend:

Variable Name	What it contains	Required?
GOOGLE_APPLICATION_CREDENTIALS_JSON	This is the full JSON key required to connect to the Firebase database. This is the replacement of the serviceAccountKey.json for a production environment.	Yes

Admin Portal Frontend:

Variable Name	What it contains	Required?
VITE_API_URL	The fulkl HTTPS URL for the backend WebService for the admin portal.	Yes

Admin Portal Backend:

Variable Name	What it contains	Required?
SERVICE_KEY	This is the full JSON key required to connect to the Firebase database. This is the replacement of the serviceAccountKey.json for a production environment.	Yes
LOGIN_CREDS	The full JSON contents of the admin credentials to login to the admin portal.	Yes

5.3 Rotating the Firebase Service Account Key

Should the service account key ever be exposed to the public, follow these instructions to retrieve a new one:

1. Open the Firebase console
2. Navigate to Project settings then to the Service Accounts tab
3. Click “Generate new private key” which will download the new JSON key
4. In Render, update the environment variables
5. Trigger a new redeploy for the backend services

5.4 Updating the Admin Portal Credentials

1. Navigate to the admin portal backend resource on Render
2. Click on the “Environment” tab and click “Edit”
3. Update the JSON object to have a new username and password
4. Click “Save Changes”
5. Trigger a new redeploy for the admin portal backend resource

6. Quick Reference

Deployment Checklist

Run through this checklist each time you push an update to the code:

1. Code changes are committed to the main branch in the repo.
2. Render Events tab shows the build status as “Building”, if not trigger a manual redeployment.
3. Watch the build log and status, if it shows as “Failed”, check the build log.
4. Once the status shows as live, open the URL and verify the change was applied.